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Task01: Project EDA Report

Submitted by

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Related work and research gap

Paper details	Dataset	Application area	Method	Results	Strengths and limitations	Research gap and relation to our work
Binary and Multi-Class Malware Threads Classification Ismail Taha Ahmed; Norziana Jamil; Marina Md. Din; Baraa Tareq Hammad **Year: 2022**	MaleVis: 14,226 RGB byte-images; 26 classes (25 malware families + 1 cleanware); 9,100 training and 5,126 testing images	Static, image-based malware detection and family classification.	RGB-to-grayscale conversion; handcrafted SFTA and Gabor texture features; Gaussian Discriminant Analysis (GDA) and Naive Bayes; 10-fold cross-validation.	SFTA-GDA: 97% binary detection accuracy and about 98% multiclass accuracy. Gabor-GDA 93%; SFTA-NB 84%; Gabor-NB 83% for binary detection	**Strengths:** Low-cost feature vectors; supports binary and multiclass tasks, useful for classical-ML baseline. **Limitations:** One dataset only. Manual feature engineering and no end-to-end learning or behavioral features.	**Gap:** Earlier texture methods used large, costly feature vectors and struggled with similar families. **Relation**: Direct MaleVis baseline for comparing our CNN model on accuracy, automation, and complexity.
A Novel Detection and Multi-Classification Approach for IoT-Malware Using Random Forest Voting of Fine-Tuning CNNs Safa Ben Atitallah; Maha Driss; Iman Almomani **Year: 2022**	MaleVis: 14,226 RGB images; 25 malware classes + 1 benign class; 70% training, 20% validation, 10% testing.	Vision-based IoT malware detection and multiclass family identification.	Transfer learning with ResNet18, MobileNetV2, and DenseNet161; hard voting, soft voting, stacking, and Random Forest-based voting; 5-fold cross-validation.	RF-voting: accuracy 98.68%, precision 98.74%, recall 98.67%, specificity 98.79%, F1 98.70%, MCC 98.65%; 672 ms average processing time.	**Strengths**: Strong ensemble benchmark, broad metrics, transfer learning and compares four fusion strategies. **Limitations:** Single dataset; three-backbone ensemble is resource-heavy and no explainability or runtime behavior.	**Gap:** Single-CNN studies often misclassified similar families and ignored ensemble benefits. **Relation:** Directly uses MaleVis deep-learning benchmark.
Efficient and Generalized Image-Based CNN Algorithm for Multi-Class Malware Detection Yajun Liu; Hong Fan; Jianguang Zhao; Jianfang Zhang; Xinxin Yin **Year: 2024**	Malimg, Microsoft BIG2015, MaleVis, and a blended Malimg+MaleVis dataset. MaleVis has 9,100 training and 5,126 test images across 26 labels	Fast, generalized visual malware multiclass detection across heterogeneous datasets.	VBDN: malware visualization, BalancedDatasetSampler weighted sampling, image augmentation, compact custom CNN, GLCM/classical-ML baselines, and confusion-matrix analysis.	Accuracy: Malimg 94.22%; BIG2015 96.19%; MaleVis 83.22% with the heterogeneous "Other" class and 96.76% after removing it; blended dataset 91.39%	**Strengths:** Four-dataset evaluation; handles imbalance; compact CNN. Detailed class-error analysis. **Limitations:** MaleVis result depends strongly on the other class. Static images omit runtime behavior	**Gap:** Existing methods often lacked cross-dataset generalization, imbalance handling, speed, and class-level error analysis. **Relation:** Guides our sampling, augmentation, lightweight CNN, and confusion-matrix design.
An Enhancement for Image-Based Malware Classification Using Machine Learning with Low Dimension Normalized Input Images Tran The Son; Chando Lee; Hoa Le-Minh; Nauman Aslam; Vuong Cong Dat **Year: 2022**	Malimg: 9,339 images, 25 families; Malheur: 3,133 samples, 24 families. MaleVis is not used	Lightweight visual malware classification for resource-constrained and IoT devices	Horizontal image reduction from 64x64 to 32x64, 16x64, and 8x64; raw pixels fed to k-NN, SVM, and a two-layer CNN; comparison with GIST features.	Malimg: k-NN 98.11% at 8x64; SVM 98.51% at 32x64; CNN about 97.3%. Malheur CNN about 91%. Training time reduced to about 1/2, 1/4, and 1/8.	**Strengths:** Simple dimensionality reduction; retains high accuracy. Also, greatly lowers training cost; useful for low-resource hardware. **Limitations:** Not MaleVis-based.	**Gap:** Square normalization and handcrafted descriptors add unnecessary dimensions and cost. **Relation:** Does not directly realted to MaleVis but supported study that motivates image-resolution and lightweight-preprocessing that helps to understand
An Explainable Hybrid CNN-Transformer Architecture for Visual Malware Classification Mohammed Alshomrani; Aiiad Albeshri; Abdulaziz A. Alsulami; Badraddin Alturki **Year: 2025**	Training/testing: Malimg, MaleVis, VirusMNIST. External validation: Maldeb and Dumpware-10. MaleVis has 9,100 training and 5,126 validation images.	Explainable, multi-dataset visual malware classification for SOC/cloud deployment	ConvNeXt-Tiny + Swin Transformer; adaptive gated fusion, auxiliary heads, knowledge distillation, AdamW training, and Grad-CAM explanations	Hybrid average accuracy 94.04%. Malimg 99.57%; MaleVis 88.65% accuracy, 90.74% precision, 96.49% recall, 92.69% F1; VirusMNIST 93.90%; Maldeb 98%; Dumpware-10 97%.	**Strengths:** Combines local and global features; XAI and multi-dataset testing; external validation. **Limitations:** High computational cost and image conversion overhead. MaleVis accuracy trails just simpler baselines.	**Gap:** Few studies combine CNNs and Transformers while remaining explainable and testing across several datasets. **Relation:** Closest modern reference for our MaleVis work and a basis for improving efficiency and MaleVis accuracy

1.Introduction

Malware is harmful software created to damage computers, steal data, disrupt systems, or gain unauthorised access. It can affect individuals and organizations by causing financial loss, privacy breaches, and system failure.

In image-based malware analysis, malware binary files are converted into images by mapping their byte values into pixel intensities. These images often contain visual patterns, textures, and structures that can be analyzed using computer vision and deep-learning models.

Malware classification is performed to identify the family or type of malware automatically. Accurate classification helps security systems detect threats faster, understand malware behaviour , and apply suitable protection measures.

2.About Dataset

The MaleVis dataset is a multi-class malware image dataset containing 14,226 RGB PNG images from 26 malware classes. The dataset is divided into 9,100 training images and 5,126 validation images, with each image having a uniform resolution of 300 × 300 pixels.

3.Objectives of the EDA

- Examine the overall structure and suitability of the MaleVis dataset for model development.
- Analyses class distribution, class imbalance, and representative malware image patterns.
- Investigate image resolution, RGB pixel characteristics, brightness, and consistency.
- Identify duplicate samples and possible train–validation data leakage.
- Use static and interactive visualizations to explore class-level and pixel-level patterns.

4.Dataset Representation

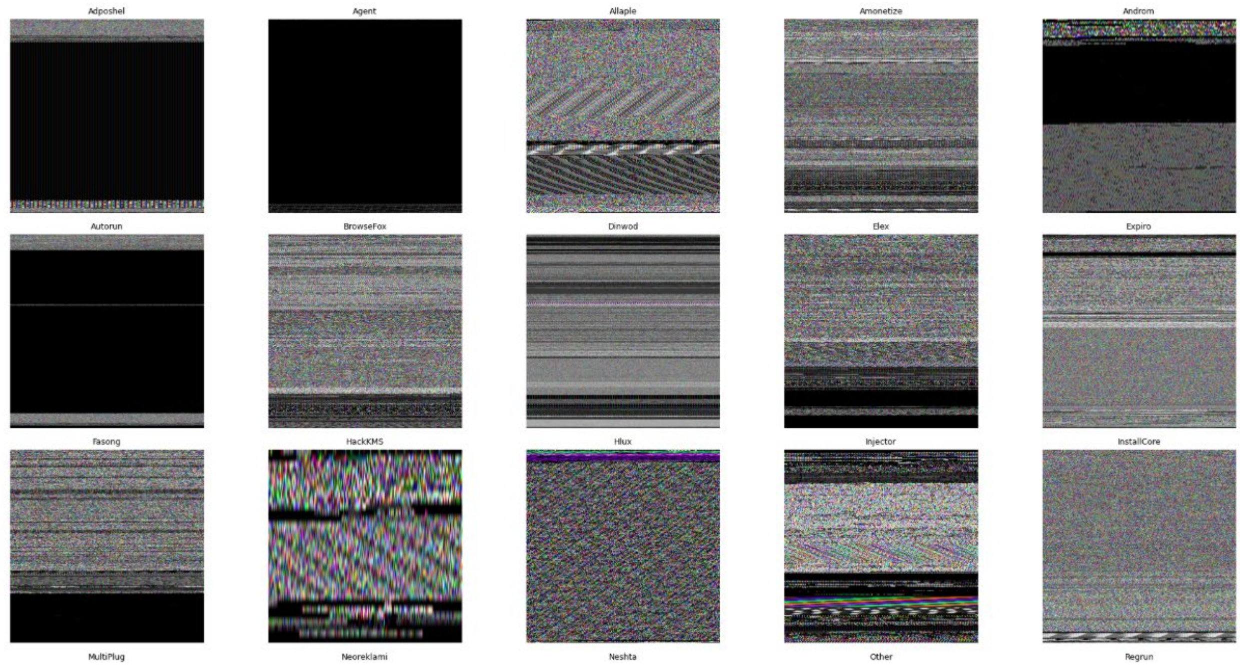

Figure 1. Representative MaleVis images

The malware images show noticeable differences in texture and structure, including horizontal bands, dark regions, repeated patterns, and dense coloured noise. Since these images are generated from malware binary files, their visual patterns represent byte-level structures rather than natural objects. These differences suggest that image-based deep-learning models can learn useful spatial and texture features for malware classification. However, some classes appear visually similar, which may increase the risk of misclassification. Therefore, class-wise performance and the confusion matrix should be examined carefully during model evaluation.

4.1 Dataset summary table

Property	Value
Dataset Name	MaleVis
Dataset source	MaleVis Dataset, Kaggle (**URL:** https://www.kaggle.com/datasets/sohamkumar1703/malevis-dataset)
Data representation	Malware binaries converted into images
Problem Type	Multi-class Malware Image Classification
Number of Classes	26
Most Populated Class	Other
Least Populated Class	Agent
Training Images	9100
Validation Images	5126
Total Images	14226
Colour Mode	RGB
Image Format	PNG
Image Resolution	300 × 300 pixels
Corrupted images	0

Property	Value
Average Images per Class	547.15

4.2 Class Count Table

Class	Train Images	Validation Images	Total Images
Adposhel	350	144	494
Agent	350	120	470
Allaple	350	128	478
Amonetize	350	147	497
Androm	350	150	500
Autorun	350	146	496
BrowseFox	350	143	493
Dinwod	350	149	499
Elex	350	150	500
Expiro	350	151	501
Fasong	350	150	500
HackKMS	350	149	499

Class	Train Images	Validation Images	Total Images
Hlux	350	150	500
Injector	350	145	495
InstallCore	350	150	500
MultiPlug	350	149	499
Neoreklami	350	150	500
Neshta	350	147	497
Other	350	1482	1832
Regrun	350	135	485
Sality	350	149	499
Snarasite	350	150	500
Stantinko	350	150	500
VBA	350	150	500
VBKrypt	350	146	496
Vilsel	350	146	496

Calculate the ratio using the total image counts:

- Majority class: Other = 1,832 images
- Minority class: Agent = 470 images

$$\text{Majority-to-minority ratio} = \frac{1832}{470} = 3.90{:}\,1$$

5. Class Distribution Graph

5.1 Overall Class-distribution

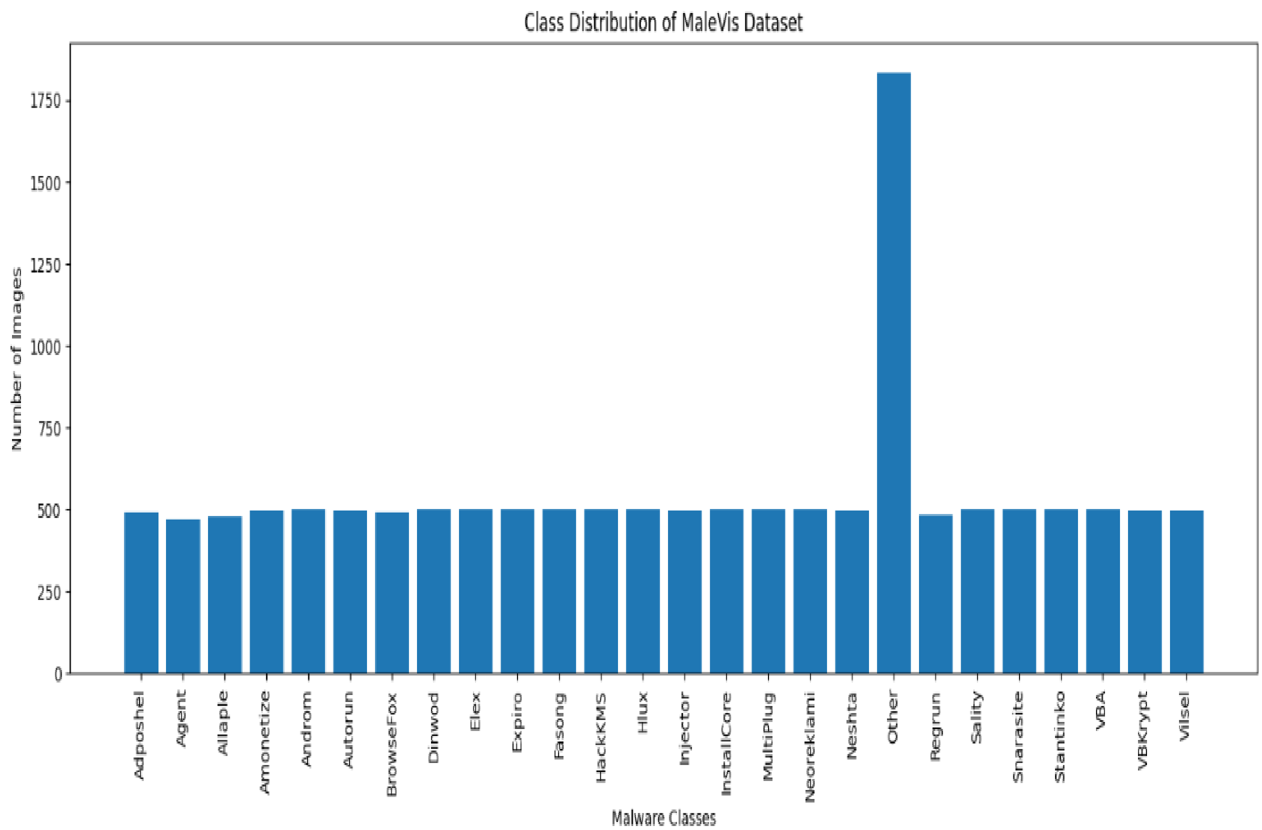

Figure 2. Overall class distribution

The overall class-distribution graph shows that most malware classes contain approximately 470–501 images. However, the Other class contains 1,832 images, making it much larger than the remaining classes. The minority class is Agent, with 470 images. Therefore, the combined dataset is imbalanced, with a majority-to-minority ratio of approximately 3.90:1.

5.2 Training and Validation class distribution

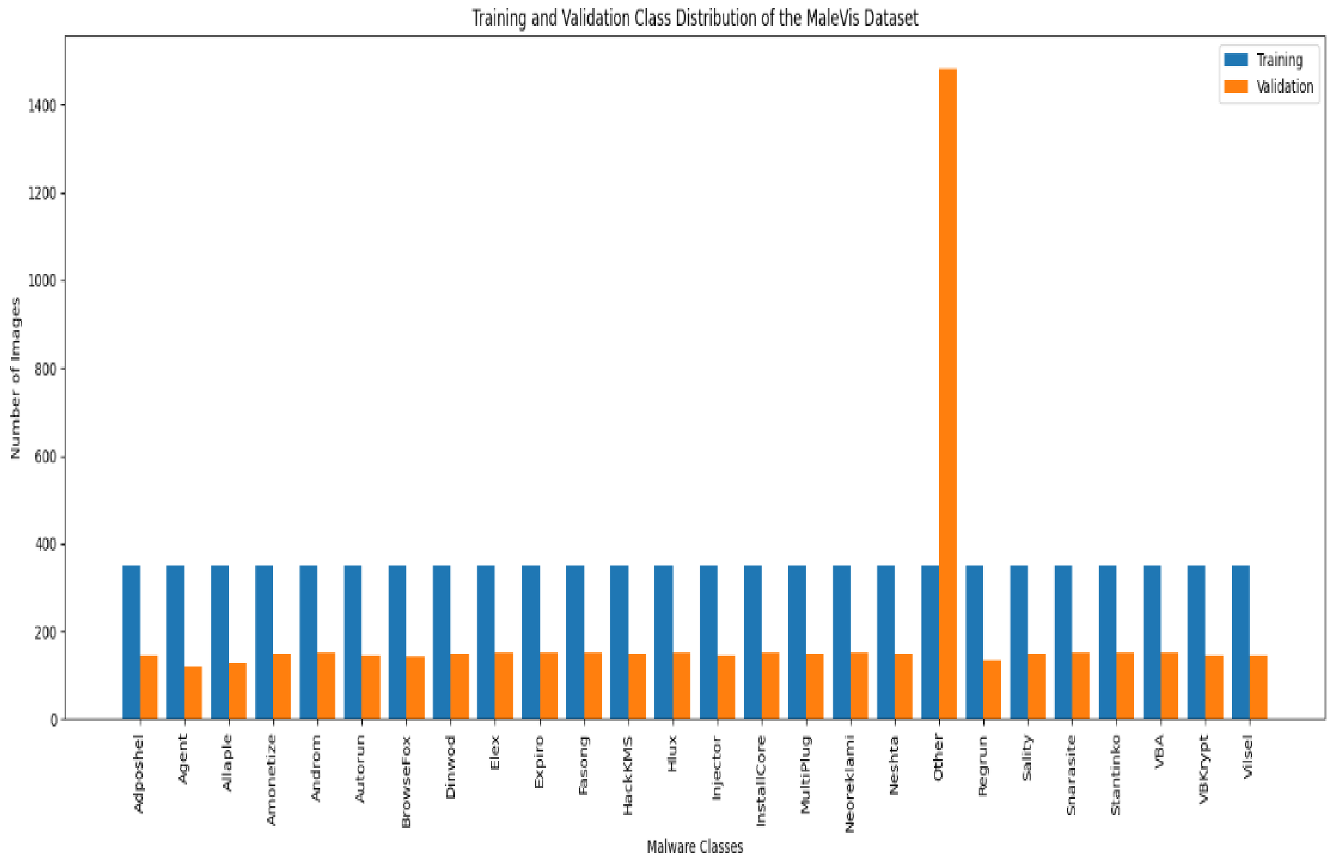

Figure 3. Training and validation class distribution

The training dataset is perfectly balanced because each of the 26 classes contains 350 images. In contrast, the validation dataset is imbalanced. Most validation classes contain approximately 120–151 images, while the Other class contains 1,482 images. This shows that the overall class imbalance mainly comes from the validation set.

6.Resolution Analysis

Figure 4. Image Resolution distribution

All 14,226 images in the MaleVis dataset have the same resolution of 300 × 300 pixels. The minimum, maximum, and average width are all 300 pixels, and the minimum, maximum, and average height are also 300 pixels. No image with a different resolution was detected, confirming that the dataset is fully standardised across all malware classes.

7.Pixel and Colour Analysis

7.1 RGB descriptive-statistics table:

Channel	Mean	Median	Standard Deviation	Minimum	25th Percentile	75th Percentile	Maximum
Red	104.23	104.0	72.54	0	47.0	151.0	255
Green	104.17	104.0	72.63	0	46.0	151.0	255
Blue	104.44	104.0	72.55	0	47.0	151.0	255

Observation:

- The mean pixel intensities of the red, green, and blue channels are 104.23, 104.17, and 104.44, respectively.
- The median pixel intensity is 104 for all three channels.
- The standard deviation is approximately 72.6, indicating a wide spread of pixel-intensity values.
- The minimum intensity is 0, and the maximum intensity is 255 for every channel.
- The RGB channels have very similar descriptive statistics, indicating comparable overall intensity distributions.

7.2 Skewness and kurtosis:

Channel	Skewness	Kurtosis
Red	0.296	-0.677
Green	0.295	-0.683
Blue	0.293	-0.680

Distribution Observation

The red, green, and blue channels have nearly identical skewness and kurtosis values. This indicates that all three channels follow similar pixel-intensity distribution shapes, with slight positive skewness and relatively flat distributions.

Analysis

The similarity in these statistics suggests that no RGB channel behaves very differently from the others in terms of distribution shape. Therefore, retaining all three channels is reasonable, as each may contribute complementary spatial and texture information to the CNN model.

7.3 RGB Histogram

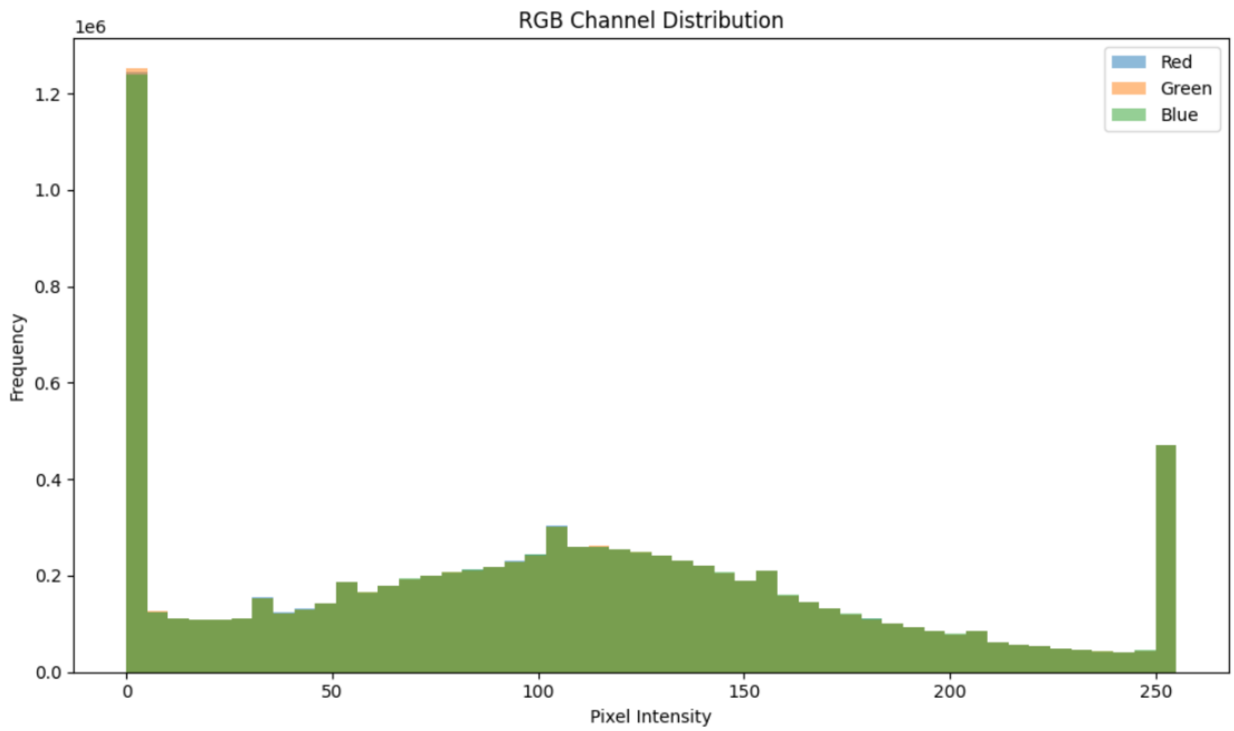

Figure 5. RGB intensity histogram

Observation

The RGB histogram shows a large concentration of pixels near intensity 0, indicating that dark regions are common in the images. Another broad concentration appears around the middle intensity range, approximately 100–150, while a smaller peak is visible near intensity 255. The three channel distributions overlap strongly.

Analysis

The histogram suggests that MaleVis images contain a mixture of dark regions, medium-intensity textures, and bright structural areas. The strong overlap between the red, green, and blue distributions confirms that no channel strongly dominates the dataset.

7.4 Violin Plot Observation

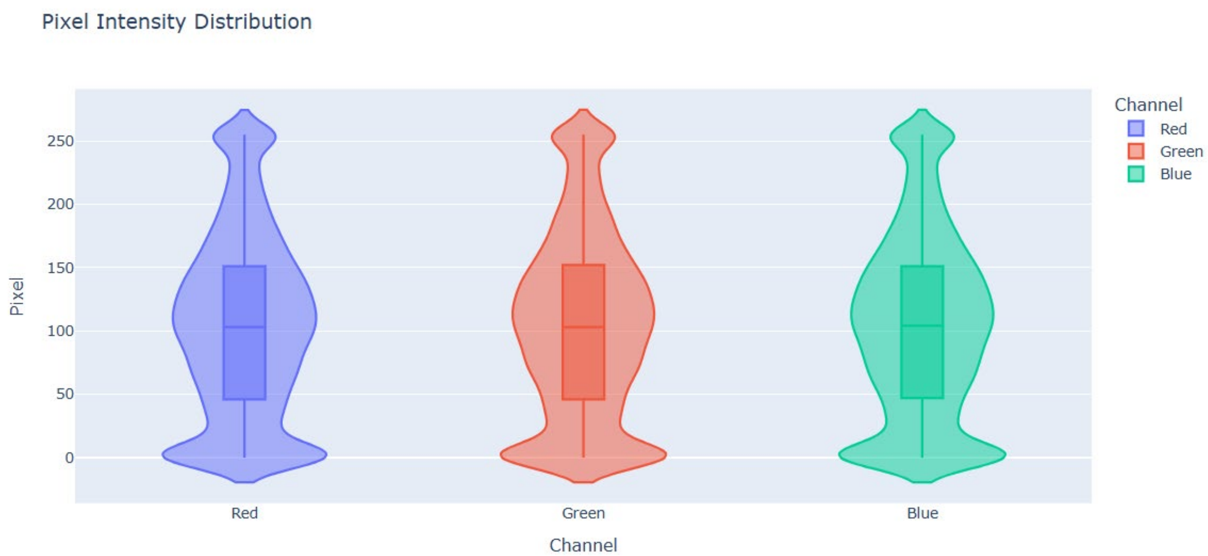

Figure 6. RGB Violin Plot

Observation

The violin plots for the red, green, and blue channels have very similar shapes, widths, medians, and spreads. Each channel contains a broad range of pixel intensities, with higher density around low and middle intensity values.

Analysis

The close similarity among the three violin plots supports the findings from the boxplot and histogram. It indicates that the RGB channels follow comparable distributions and that useful information is likely distributed across all three channels rather than concentrated in only one channel.

7.5.RGB Box Plot

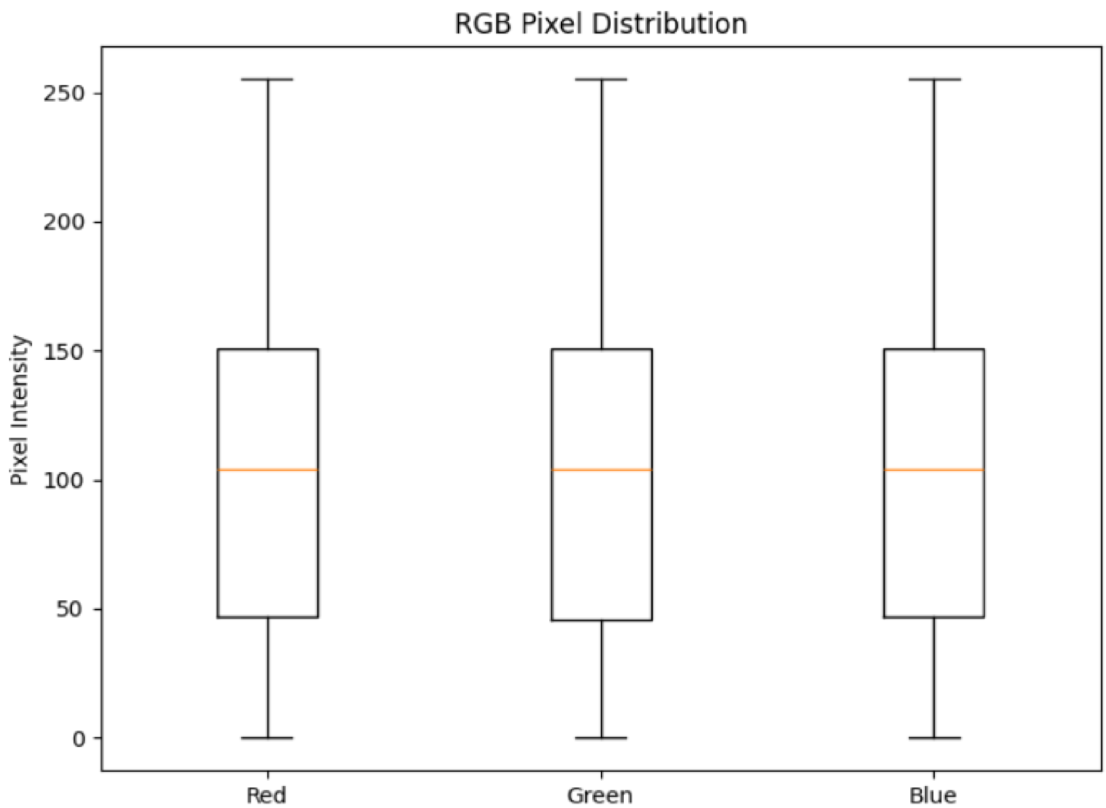

Figure 7. RGB Box Plot

Observation

The boxplot shows that the red, green, and blue channels have almost identical median pixel-intensity values. Their interquartile ranges are also very similar, and all three channels extend across the full intensity range from 0 to 255.

Analysis

The similar medians and spreads indicate that no individual color channel has a noticeably different intensity pattern. This suggests that the RGB channels contribute comparable amounts of information to the malware-image representation.

8.Image Quality Analysis

8.1. Brightness quality

Minimum brightness: 21.51

Maximum brightness: 226.81

Mean brightness: 104.28

Median brightness: 105.19

Brightness standard deviation: 34.66

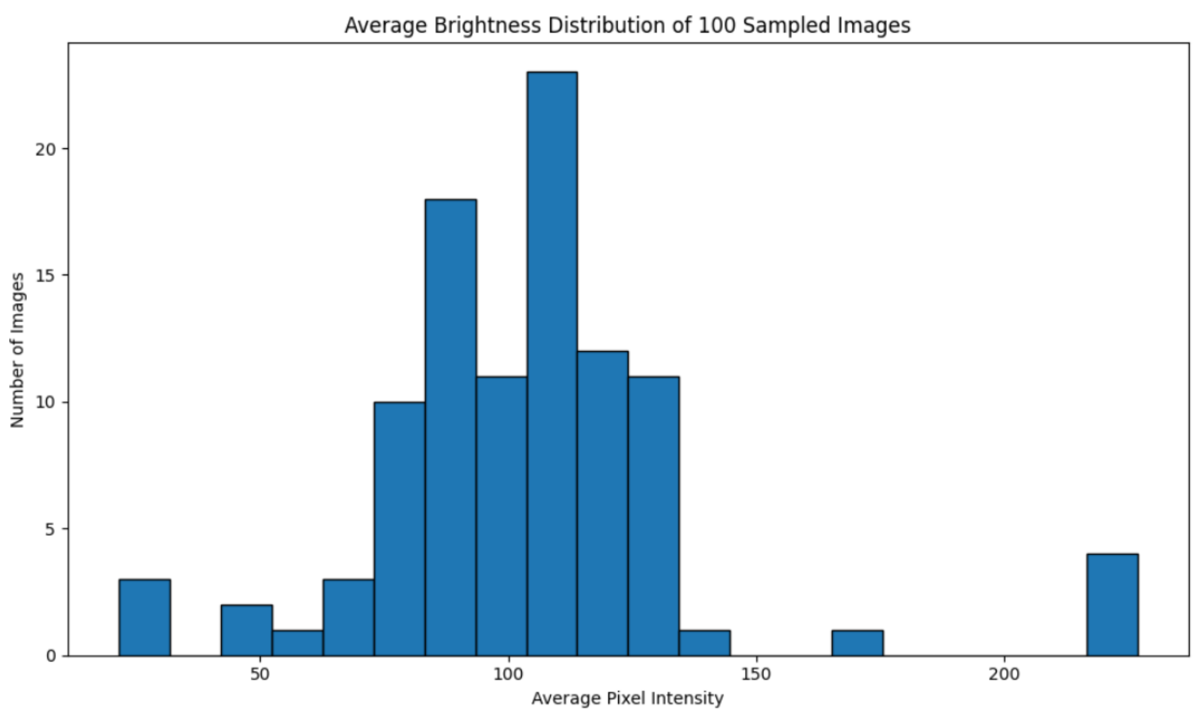

Figure 8. Brightness distribution

The average brightness of the 100 sampled images ranges from 21.51 to 226.81. Most images are concentrated approximately between pixel intensities 80 and 130. The mean brightness is 104.28, while the median is 105.19, showing that the center of the distribution is around a moderate brightness level. A few images are considerably darker or brighter than the majority.

Analysis

The brightness standard deviation of 34.66 indicates noticeable variation among the sampled malware images. This variation is likely related to differences in byte-derived textures, dark bands, and bright structural regions rather than photographic lighting conditions. The extremely dark and bright samples should be visually inspected, but they should not be treated as poor-quality images automatically because these intensity patterns may contain useful information for malware classification.

8.2.Contrast analysis

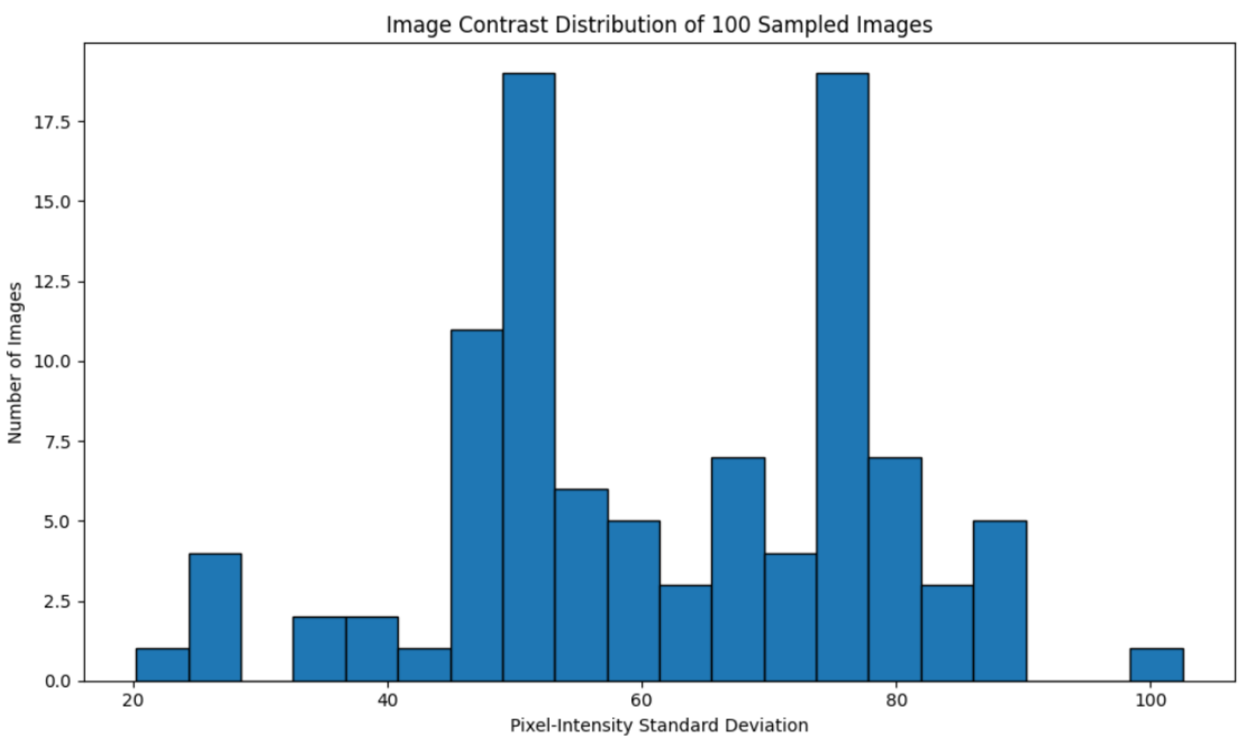

Figure 9. Image Contrast distribution

Minimum contrast: 20.22

Maximum contrast: 102.51

Mean contrast: 61.68

Median contrast: 59.12

The contrast values of the 100 sampled images range from 20.22 to 102.51, with a mean of 61.68 and a median of 59.12. Most images fall within a moderate contrast range, although the histogram shows noticeable groups around approximately 45–55 and 70–80. A small number of images have unusually low or high contrast.

Analysis

Contrast was measured using the standard deviation of pixel intensities. Images with higher contrast contain stronger differences between dark and bright regions, while lower-contrast images contain more uniform or less varied textures. In the MaleVis dataset, these differences may reflect byte-derived structures such as horizontal bands, dark blocks, repeated patterns, and dense noise-like regions. CNN-based models may use these local contrast and texture characteristics to distinguish between malware classes.

9.Data Quality Analysis

A comprehensive data-quality assessment was conducted to evaluate the completeness, consistency, integrity, and reliability of the MaleVis dataset before model development.

9.1. Dataset Completeness and File Integrity

No missing or unreadable images were identified. All image files were loaded successfully, indicating that the dataset does not contain missing values in the form of absent, inaccessible, or corrupted image files.

All 14,226 images were confirmed to be:

- Readable PNG files
- RGB color images
- Uniformly sized at 300 × 300 pixels
- Correctly organized within their respective class folders

No missing classes were found in either the training or validation split.

9.2. Dataset Consistency

The dataset was structurally consistent across all major quality dimensions. No unexpected image resolutions, non-RGB images, corrupted files, or class-folder inconsistencies were detected.

Quality check	Result
Total images	14,226
Corrupted images	0
Missing training classes	0
Missing validation classes	0
Unexpected image resolutions	0
Non-RGB images	0
Exact duplicate groups	187
Images involved in duplicate groups	480
Validation images duplicated in training	173

These findings indicate that the dataset is generally clean and consistent in terms of file format, image dimensions, colour mode, readability, and class organization.

9.3. Duplicate Image Analysis

A total of 187 exact duplicate groups were identified, involving 480 image files. Among these, 173 validation images had exact matches in the training set. The remaining duplicate images occurred within the same dataset split.

The presence of identical images in both the training and validation sets creates a potential data-leakage problem. A model may encounter an image during training and later be evaluated on the same image in the validation set. Consequently, validation accuracy and other performance metrics may be artificially inflated and may not accurately represent the model's ability to generalize to unseen malware samples.

Therefore, cross-split duplicates should be removed before final model training and evaluation. A safer approach is to group identical images using their file hashes and then create a new stratified train-validation-test split, ensuring that all images belonging to the same duplicate group remain within a single split.

9.4. Class-Imbalance Analysis

Although the combined dataset is moderately imbalanced, the supplied training split is perfectly balanced, with 350 images in each of the 26 categories. Therefore, class weighting or balanced sampling is not required for the initial baseline models. However, the validation split is highly imbalanced because the Other category contains 1,482 images. Consequently, validation accuracy may be strongly influenced by performance on this category. Macro-F1, per-class precision, per-class recall, weighted-F1 and the confusion matrix should therefore be reported. If the dataset is deduplicated and re-split, class balance should be recalculated before selecting an imbalance-handling technique.

10.Final EDA Findings

- The MaleVis dataset contains 14,226 RGB malware images distributed across 26 classes.
- The training split contains 9,100 images and is balanced, with 350 images per class.
- The validation split contains 5,126 images and is imbalanced, mainly because the Other class contains substantially more images than the remaining classes.
- The majority class is Other with 1,832 images, while the minority class is Agent with 470 images, producing an imbalance ratio of approximately 3.90:1.
- All images are readable PNG files with a consistent resolution of 300 × 300 pixels and RGB color mode.
- No corrupted images, missing classes, inconsistent resolutions, or non-RGB images were detected.
- The sample images show class-dependent differences in texture, colour, banding, and structural patterns.

- The RGB channels have similar mean, median, spread, skewness, and kurtosis values.
- The RGB histograms, boxplots, and violin plots indicate that the three channels have broadly comparable intensity distributions.
- Brightness and contrast vary across images, indicating differences in malware byte-pattern structures.
- A total of 187 exact duplicate groups and 480 additional duplicate files were identified.
- A total of 173 validation images also occur in the training set, creating a potential data-leakage risk.
- Duplicate handling and careful treatment of class imbalance should be performed before fair model training and evaluation.
- Overall, the dataset is suitable for CNN and CNN-attention experiments after addressing these issues.